

with past and present information related to the IoT application at hand. A sensor could be added to this RFID tag for real-time monitoring and information gathering. The information thus gathered from numerous sources is then sent to a cloud for processing, which completes the IoT. The cloud and the IoT go hand-in-hand with their common background of distributed processing.

Before the gathered information is sent to the network managing the other 'things' of the IoT, there is RFID middleware, which contains the logic of the application and a back-end database that, for instance, runs on MySQL and is used to store data. It allows the data captured by the devices to be routed to another system, which is ultimately uploaded and utilised by the IoT application.

Forrester Research defines RFID middleware as "platforms for managing RFID data and routing it between tag readers or other auto identification devices and enterprise systems."

One of the significant aspects here is to transform low-level data gathered from tags into high-level data that can drive information and enable the IoT application to perform accurately. So it is extremely important that metadata is defined for all tags and the antennae that monitor them.

The metadata stored on tags is recorded, and over a period of time

How IoT has revolutionised devices

1. Traffic cameras and sensors in smartphones have enabled us to predict peak traffic conditions. The data is gathered from users, processed and sent back to users as useful information. This allows people to bypass traffic based on real-time feedback, forming the basis for an intelligent transportation system.
2. Integrating home lighting, ventilation and security systems with wireless connectivity has allowed people to control these systems from a remote location by using a mobile device. In fact, home appliances like washing machines, refrigerators and entertainment systems can also be remotely monitored and controlled. In the future, this could evolve to create a fully automated society.
3. Microsensors on human bodies or objects that regularly come into contact with the human body can be used to sense and monitor the health of individuals non-invasively. This technology can be taken ahead to form an e-health initiative where doctors can monitor their patients from a remote location in real time.
4. Logistics at major retailer or transportation firms can be automated by using IoT-enabled kiosks for all arriving and departing objects. Objects will be tagged with metadata pointing to location and other details and stored in a virtual object that can then be accessed remotely through a network. The system will gather all the required location data and reroute objects through the most efficient path.
5. The futuristic merging of cutting-edge robotics and new sensor systems along with growth of the Internet of Things could herald embedded systems into a new age of cyber-physical systems.

used to understand 'events.' Events are created based on what the IoT application is designed for. For instance, in an office, people tagged with RFID tags would allow the system to understand and record events like "Rahul is in the office" or "There are eight people in the meeting hall."

The abilities of the IoT can also be used to create a search engine of real-world things. Utilising location data gathered from the tags, users would be able to locate a person or personal item by querying the database for its last known location.

Apps, apps and more apps

RFIDs are not the only means. Current smartphones, tablets, and even the Nike sports band that monitors your heartbeat rate and blood pressure are all part of the Internet of things. These gather data from you, and transmit it to the Internet through specific apps, where a central server processes the data and converts it into meaningful information.

From what we have learnt from the smartphone boom, apps are the most important element for a product to be successful in the market. Therefore it is extremely important for you to have the ability to rapidly create IoT applications. One such approach favoured by the Media and Graphics lab at the University of British Columbia focuses on a lightweight toolkit for developing IoT applications. It targets rapid development using Web technologies and protocols and has been described at the 2012 IoT conference by IEEE.

"WoTKit is a Java Web application that leverages the Spring Framework7—a popular development framework for enterprise applications. The data model consists of sensors and time-stamped sensor data. The Apache Active MQ message broker8

Some components designed for IoT

1. **NXP JenNet-IP-EK040 evaluation kit.** This kit provides all the components needed to create lighting and home automation applications for IPv6-based networks. It is based on NXP's JN514x series of ultra-low-power single-chip wireless microcontrollers.
2. **Renesas R32C MCU. R32C/100 Series** is upwardly compatible with M32C/80 and supports key features of M16C family, including low power consumption, low noise emissions and high EM/EMS performance.
3. **Freescal TWR-METRO-KIT.** This kit enables you to build a 6LoWPAN-compliant smart object network out of the box and integrate it with your own hardware for network performance evaluation in a real environment for the Internet of Things or machine-to-machine applications.
4. **TI CC-BLOWPAN-DK-868 development kit.** This kit supports 868MHz frequency band and can quickly be reconfigured in software for operation in the 915MHz ISM band. Additional evaluation module kits can be purchased separately to support other frequencies.
5. **Higgs-3 IC.** Higgs-3 is a highly integrated, 800-bit memory, single-chip UHF RFID tag IC. It conforms to the EPC global Class 1 Gen 2 specifications.